

Apoorv Patidar

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Professional Summary

Computer Science undergraduate building scalable AI systems, with hands-on experience in LLM-powered applications, RAG pipelines, and production-grade ML models. Skilled in MLOps, cloud deployment (AWS), and containerization (Docker). Proficient in Python, TensorFlow, PyTorch, and LangChain for developing and deploying intelligent systems end-to-end

Technical Skills

Programming Languages: Python, C++, Swift

MLOps and Cloud: AWS (EC2, CodePipelines, ECS, ECR, Elastic Beanstalk), Docker, GitHub Actions, CI/CD, Git, DagsHub

Web Frameworks and Databases: Flask, FastAPI, Streamlit, MySQL, PostgreSQL, MongoDB

AI and ML: Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, CrewAI, MLflow, Large Language Models (LLMs)

Projects

FusionMind RAG — Multimodal, grounded reasoning for your docs and images

- Engineered a multimodal RAG pipeline that fuses PDF text, CLIP-driven image signals, and Gemini (via LangChain) to deliver grounded, source-cited answers across text/PDF/image inputs.
- Built a low-latency retrieval layer (PDF cleaning, recursive chunking, Google/HF embeddings with smart fallback, FAISS with MMR) improving context relevance by 10% while also reducing query time.
- Orchestrated label-aware query expansion by auto-labeling uploaded images and merging DuckDuckGo web snippets as auxiliary context, reducing off-topic generations while improving answer relevance by 20%.
- Delivered a production-ready Flask UI/API with per-session chat history, attachment management, warm-start model loading, and quota-aware failovers, hardening reliability and operational readiness.

Automated Academic Peer Review Assistant

- Built a Flask backend to parse academic PDFs and generate structured reviews, reducing evaluation time by 60%.
- Implemented a FAISS retrieval system and optimized research paper searches by topic, which reduced search retrieval time by 40%.
- Designed Python scripts for citation extraction, fact checking, novelty detection, and plagiarism check, consolidating results into a single review file for streamlined analysis.
- Engineered a React + TypeScript frontend and integrated it with a Flask API to enable seamless PDF uploads.

FloraNet - CNN Model for Multi-Class Flower Classification

- Developed a convolutional neural network using TensorFlow/Keras to classify five flower species, achieving 81% validation accuracy with EfficientNetB0 architecture.
- Improved accuracy by 8.5% through transfer learning and hyperparameter tuning.
- Optimized training speed via data augmentation (rotation, scaling) and preprocessing, reducing validation loss by 35%.
- Conducted comparative performance analysis of CNN, VGG16, and EfficientNetB0 architectures with detailed hyperparameter reports.

Education

Vellore Institute of Technology, Bhopal

• B.Tech - Computer Science and Engineering - **CGPA: 8.6/10.0**

Oct 2022 – Oct 2026

• **Specialization:** Artificial Intelligence and Machine Learning

• **Coursework:** Operating Systems, Computer Networks, Database Management Systems (DBMS), Artificial Intelligence

Certifications

- OCI Generative AI Professional - Oracle (Aug 2025)
- Introduction to Model Context Protocol - Anthropic (Aug 2025)
- Gen AI Intensive Course - Kaggle (Mar 2025)

Languages and Hobbies

Languages: English (Professional Fluency), Hindi (Native), Spanish (Basic Fluency)

Hobbies: Gaming, Football